Delta-Like Ligand 4 Binding Promotes Presenilin Cleavage and HES1 Expression in Arteries

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DOI: 10.1161/ATVBAHA.124.32111 | PMID: 39231011 | PMCID: PMC10923011

Page 1 — Abstract & Introduction

Vascular morphogenesis requires strictly regulated cell fate decisions between tip and stalk endothelial cells.

Page 2 — Receptor Cleavage

Binding of DLL4 to Notch1 induces gamma-secretase cleavage, releasing the active Notch intracellular domain (NICD).

Page 3 — Nuclear Translocation

Liberated NICD associates with RBP-Jkappa in the nucleus to stimulate transcription of downstream repressor genes.

Page 4 — Results: Target Activation

The NICD-RBP-Jkappa complex directly upregulates HES1 expression to inhibit VEGFR2 and suppress endothelial sprouting.

Page 5 — Discussion

Our findings show that DLL4 promotes HES1 expression through NICD release to regulate arterial caliber. Vehicle control did not stimulate HES1.